

Nonlinear Model Predictive Control based on multi-scale models: is it worth the complexity?

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Abstract: Nonlinear model predictive control (NMPC) has been used to control a variety of bioprocesses. As NMPC requires full state information, a nonlinear moving horizon estimator (NMHE) is used to reconstruct the states from the available measurements. In context of bioprocesses, both NMPC and NMHE typically use macroscopic mass balance type models. These models ignore all intracellular information that is available via, e.g., Genome-Scale Metabolic networks. However, the use of these models adds significantly to the computational complexity of the controller. In this paper, we aim to assess whether incorporating metabolic networks in an NMPC framework leads to improvement in the tracking performance of a continuous bioreactor. To avoid complex bilevel optimisation problems, our NMPC scheme utilises the dynamic Metabolic Flux Analysis framework. In this framework, the intracellular fluxes are estimated from the measurements of extracellular metabolites. This is achieved by an NMHE scheme. The tracking performance of this metabolic network-based NMPC-NMHE is compared to a simple macroscopic model based NMPC in which all the states, i.e., the extracellular metabolites are assumed to be measured. In our case study, the tracking performance of the metabolic network-based NMPC is marginally better than the performance of the macroscopic model based NMPC. The added advantage of metabolic network based NMPC comes from the insight they offer from the estimated fluxes. Moreover, the performance and the insights obtained depend on the complexity of the metabolic network used.

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1. INTRODUCTION

Nonlinear Model predictive control (NMPC) of bioprocesses utilises a model-based prediction to devise a dynamic input profile that optimises a certain objective (e.g., yield, productivity) over a certain period (called the prediction horizon). However, instead of applying the entire input profile to the process, only a part of it is applied until the next measurement is available. Once the measurement is available, the model predictions are optimised again over a shifted prediction horizon; this time using the new measurements as the initial values. This incorporates a feedback mechanism in the control structure that can handle model-process mismatch and other disturbances. NMPC has found widespread applicability in a variety of engineering fields such as autonomous vehicles (Farag, 2021), airborne wind energy management (Vukov et al., 2015), as well as bioprocess control (Lucia and Engell, 2013; Abadli et al., 2022). Full state information (i.e., measurement of all the modelled variables) essential for NMPC, is rarely available in practice. Thus, the unknown states need to be reconstructed from the measured states. The estimator equivalent of the NMPC, the Nonlinear Moving Horizon Estimator (NMHE), utilises the measurements over a certain period in the past (called the estimation horizon) in the least squares optimisation

framework to estimate the unknown states. The combined NMHE-NMPC approach is also widespread in a variety of engineering fields (Vukov et al., 2015; Huang et al., 2021, e.g.).

Model-based control of bioprocesses is mostly based on macroscopic mass balance models. These models quantify the dynamics of macroscopic entities such as biomass, substrate, and product, but ignore intracellular dynamics. This cellular information on the fluxome, the metabolome, the proteome, the transcriptome, and the genome is available through the Genome-Scale Metabolic networks (GEMs). On a metabolic level, the cell behaviour is described by various metabolites, their conversion, interactions, and transport between intracellular and extracellular space. The stoichiometric matrix $\mathbf{S}$, with each of its rows corresponding to a metabolite and columns to a reaction, represents these relationships. As intracellular dynamics are expected to be much faster than extracellular dynamics, a pseudo-steady state for the intracellular metabolites can be assumed. However, as the number of measured fluxes is still fewer than the total number of fluxes, the network system is underdetermined. For that reason, constraint-based modelling approaches are used to establish the entire flux distribution. Two approaches are commonly used. In *Flux Balance Analysis* (FBA), it is assumed that the cellular behaviour is designed to optimise

a certain cellular objective (e.g., cell growth) and the fluxes are determined by solving an optimisation problem that mathematically describes the cellular objective. Any flux measurements available are then added as constraints. In *Metabolic Flux Analysis* (MFA), the flux distribution is obtained by solving a constrained parameter estimation problem using available measurements in a reduced flux space.

Recently, the control of bioprocesses using metabolic network-based NMPC has gained traction. Chang et al. (2016) use the dynamic FBA (dFBA) based NMPC for closed-loop optimisation of *Saccharomyces cerevisiae* fermentation. They demonstrate an 8-14% increase in ethanol production using the closed-loop NMPC compared to an open-loop dynamic optimisation. However, solving the bilevel optimisation problems posed by the incorporation of dFBA in the optimisation framework is not trivial. de Oliveira et al. (2021) overcome this challenge by replacing the inner FBA optimisation problem with a surrogate model. The surrogate model is built by a multitude of a priori dFBA-only simulations under a variety of conditions. However, as surrogate models are completely black-box, they can only operate within a set of conditions they are trained for. Recently, we demonstrated the utility of a dMFA based NMPC (Nimmegeers et al., 2021). The intracellular flux distribution is estimated using an NMHE framework as described in Vercammen et al. (2016).

In all cases, the incorporation of metabolic information leads to an increase in computational complexity. This begs the question: is the increase in the computational cost justified by an improvement in the controller performance? In this paper, we aim to address this question through a simple toy example.

2. NUMERICAL MATERIALS AND METHODS

In this section, the toy case study is described. The *process* is assumed to be the fermentation of an organism that follows the metabolic network with predefined kinetics for the free fluxes (Section 2.3.1). Based on initial simulation data, a macroscopic model which describes the evolution of the substrates, biomass, and the product is identified (Section 2.5). This model serves as the basis for the comparison of metabolic network-based NMPC with traditional NMPC. For the metabolic network-based NMPC, all the fluxes are estimated from the extracellular measurements using an NMHE scheme described in Section 2.2. For the estimation, the fluxes are parameterised as described in Section 2.4.

2.1 Nonlinear Model Predictive Control

As mentioned, the NMPC solve an open-loop dynamic optimisation problem over the time interval $[t_k, t_{k+N_p}]$. In this paper, N_p (i.e., the prediction horizon) is set to 4. The measurements are assumed to be available every hour, thus the process is optimised over a period of 4 h. For a general process model $\dot{\mathbf{x}} = f(\cdot)$, the dynamic optimisation problem to be solved over the prediction horizon is described by

$$\min_{\mathbf{x}(\cdot), \mathbf{u}(\cdot)} J = \int_{t_k}^{t_{k+N_p}} L(\mathbf{x}(\tau), \mathbf{u}(\tau),) d\tau + M(t_{k+N_p}) \quad (1)$$

subject to:

$$\dot{\mathbf{x}}(t) = f(\mathbf{x}(t), \mathbf{u}(t)) \quad (2)$$

$$\mathbf{x}(t_k) = \hat{\mathbf{x}}_k \quad (3)$$

$$\mathbf{0} \geq \mathbf{c}(\mathbf{x}, \mathbf{u}, t). \quad (4)$$

where $\hat{\mathbf{x}}_k$ is the measurement (or the estimate) of the state $\mathbf{x}_k$ at time t_k . $\mathbf{c}(\cdot)$ represent any constraints on the states and the controls which might be applicable.

In this paper, the NMPC is used to regulate the CSTR to the required set point $\mathbf{x}_{\text{set}}$. Thus, the traditional quadratic regulatory objective described below is used.

$$J = \int_{t_k}^{t_{k+N_p}} \left(\mathbf{x}(\tau) - \mathbf{x}_{\text{set}} \right)^\top Q \left(\mathbf{x}(\tau) - \mathbf{x}_{\text{set}} \right) d\tau \quad (5)$$

2.2 Nonlinear Moving Horizon Estimation

In NMHE, a dynamic parameter estimation problem over the interval $[t_{k-N_e}, t_k]$ is solved at every time step t_k . To avoid memory overload, the measurements before the first point in this interval (t_{k-N_e}) are discarded. The NMHE is used to estimate the internal flux vector from the extracellular measurements following the formulation described by Vercammen et al. (2016). The determination of the dynamic flux vector requires the fluxes to be parameterised, i.e., $\mathbf{u} \approx \tilde{\mathbf{u}}(t, \mathbf{p})$. The parameterisation used in this paper is described further in this section. For the general model $\dot{\mathbf{x}} = f(\cdot)$, the NMHE formulation is given by

$$\min_{\substack{\mathbf{x}_{H_e}^c, \\ \mathbf{w}_{k-N_e}, \dots, \mathbf{w}_i}} \sum_{j=k-N_e}^k \|\mathbf{m}_j - \mathbf{y}(t_j)\|_{\mathbf{V}}^2 + \sum_{j=k-N_e}^k \|\mathbf{w}_j\|_{\mathbf{W}}^2 + \|\mathbf{x}_i^c - \bar{\mathbf{x}}_i^c\|_{\bar{\mathbf{P}}_i} \quad (6)$$

subject to:

$$\dot{\mathbf{x}}(t) = f(\mathbf{x}(t), \tilde{\mathbf{u}}(t, \mathbf{p}) + \mathbf{w}_x \quad (7)$$

$$\mathbf{x}(t_{i-N_e}) = \hat{\mathbf{x}}_{i-N_e} \quad (8)$$

$$\mathbf{0} \geq \mathbf{c}(\mathbf{x}, \mathbf{u}, t). \quad (9)$$

The optimisation variables are the augmented state vector $\mathbf{x}_{H_e}^c$ which contains both the state and parameter vector at t_k , and the process noise estimates w_i , $\forall i \in [k - N_e, \cdot, k]$. The arrival cost $\bar{\mathbf{x}}^c$ and $\bar{\mathbf{P}}_i$ are obtained via the unscented Kalman filter (Qu and Hahn, 2009).

2.3 Case Study: Process Model

The case study considers a continuous stirred tank reactor (CSTR) in which an organism is used to produce a product F_{ext} . The organism consumes two substrates A_{ext} and E_{ext} to produce the biomass X and the product. It is assumed that the inlet flow rate and bioreactor volume are kept constant. However, the composition of the feed can be altered. The process model is then described by

$$\frac{d\mathbf{C}_{\text{ext}}(t)}{dt} = \mathbf{S}_{\text{ext}} \cdot \mathbf{K} \cdot \mathbf{u}(t) \cdot C_x(t) + \left(\mathbf{C}_{\text{in}} \mathbf{w}(t) - \mathbf{C}_{\text{ext}}(t) \right) \cdot D. \quad (10)$$

Here, D is the dilution rate kept constant at 0.1 h^{-1} , $\mathbf{C}_{\text{ext}}$ is the vector of extracellular components (i.e., two

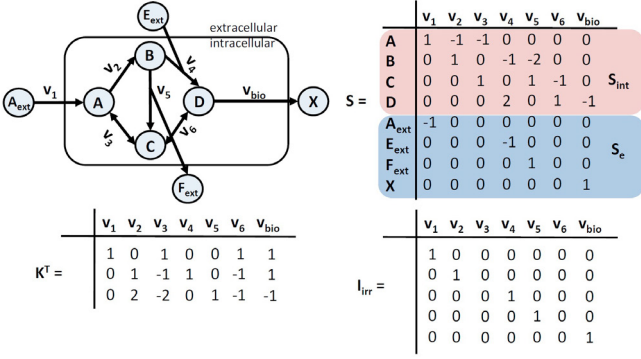


Fig. 1. Metabolic network and the characterising matrices

substrates C_A and C_E , the product C_F , and the biomass C_x , C_{in} is the substrate feed composition matrix, and $\mathbf{w}$ is the weight assigned to these feeds. The concentration of product and biomass in the feed are 0, and thus the corresponding weights are set to 0. The only manipulated variable in the system is the weight, which is constrained by $\sum_i w_i = 1$. $\mathbf{u}$ is the vector of free fluxes.

Metabolic Network The organism's metabolic network and the corresponding stoichiometric matrix are depicted in Figure 1. The network consists of three extracellular metabolites apart from biomass, four intracellular metabolites, and seven fluxes. Thus, the number of free fluxes is three. The stoichiometric matrix can be divided into an internal component (denoted by subscript int), and an external component (denoted by subscript ext). The matrix I_{irr} highlights the irreversibility of some of the reactions. $\mathbf{K}$ is the null basis of the stoichiometric matrix. Although several methods can be used to determine the null basis (as it is not unique), a fixed basis is used here.

Flux models for process simulation For the simulation of process data, the three free fluxes are assumed to follow predefined reference kinetics. These reference kinetics are

$$u_{1,ref} = \frac{C_{A,ext}}{1.5 + C_{A,ext}} \quad (11)$$

$$u_{4,ref} = 0.2 \frac{C_{E,ext}}{3 + C_{E,ext} + \frac{C_{E,ext}^2}{5}} \quad (12)$$

$$u_{5,ref} = \frac{1}{1 + C_{F,ext}}. \quad (13)$$

Using the simulation kinetics, measurements for the external metabolites can be generated. Additive, independent, Gaussian noise ($\sim \mathcal{N}(0, 0.01)$) is added to simulated measurements. In all cases, the process is considered over a period of 140 h, with a sampling frequency of 1 h^{-1} .

2.4 Flux model for estimation

A variety of free flux parameterisation strategies have been proposed in literature. In this paper, a non-linear kinetic model which can describe both the positive and negative effects of the components on reaction rate is used (Haag et al., 2005). This model, which uses one parameter per component and one maximum rate parameter, is given by

Table 1.

Parameter	Estimated Value	95% CI
$\mu_{A,max}$	0.9235	(0.7066, 1.1403)
K_A	1.3324	(0.8969, 1.7679)
$\mu_{E,max}$	0.0872	(0.0794, 0.0950)
K_E	0.8378	(0.5293, 1.1462)
$\mu_{F,max}$	0.8957	(0.8508, 0.9405)
K_F	0.2111	(-0.0375, 0.4597)

$$\tilde{u}_i(t) = u_{\max,i} \cdot \prod_{j=1}^{m_{ext}} \alpha_{ij} \cdot C_{\text{ext},j}(t) \quad (14)$$

with,

$$\alpha_{ij}(c_{\text{ext},j}(t)) = \begin{cases} \frac{C_{\text{ext},j}}{c_{\text{ext},j} + D_{ij}^2} & \text{if } D_{ij} \geq 0 \\ 1 & \text{if } D_{ij} < 0 \\ \frac{1}{1 + c_{\text{ext},j} \cdot D_{ij}^2} & \text{if } D_{ij} < 0 \end{cases} \quad (15)$$

where, $\mathbf{u}_{\max}$ is the vector of maximum rate parameters, and $\mathbf{D}$ is the matrix of modulation parameters. For the case study used in this paper, a total 12 parameters need to be estimated using the NMHE approach described earlier.

2.5 Macroscopic model

The macroscopic model which is used to described the process is give by

$$\frac{d\mathbf{C}_{\text{ext}}(t)}{dt} = \boldsymbol{\mu} \cdot \mathbf{C}_x(t) + (\mathbf{C}_{in}\mathbf{w}(t) - \mathbf{C}_{\text{ext}}(t)) \cdot \mathbf{D}. \quad (16)$$

The reaction rates $\boldsymbol{\mu}$ are assumed to follow some predefined kinetics whose parameters need to be identified. These kinetics are

$$\mu_A = -\mu_{A,max} \frac{C_{A,ext}}{K_A + C_{A,ext}} \quad (17)$$

$$\mu_E = -\mu_{E,max} \frac{C_{E,ext}}{K_E + C_{E,ext}} \quad (18)$$

$$\mu_F = \mu_{F,max} \frac{1}{K_F + C_{F,ext}} \quad (19)$$

$$\mu_X = -\sum_i \mu_i, \quad i \in \{A, E, F\}. \quad (20)$$

The choice of kinetic models is typically based on the process and is not known. In this example, the substrate uptake rates are described by Monod kinetics, and for the product a self-inhibiting kinetic model is used.

3. RESULTS AND DISCUSSIONS

The kinetic parameters described above are identified based on simulation data obtained at various conditions. Note that the experimental conditions are not optimised, and thus it could be possible to obtain better estimates. However, for this study, the parameter estimates are deemed good enough. The estimated parameters and their confidence intervals are reported in Table 1. The model performance for three different feed compositions is depicted in Figure 2. In all three cases, a slight mismatch between the macroscopic model and the metabolic network-based process can be observed.

3.1 NMPC tracking performance

In this section, the NMPC performance with the metabolic network model (NMPC-MN) is compared to the NMPC

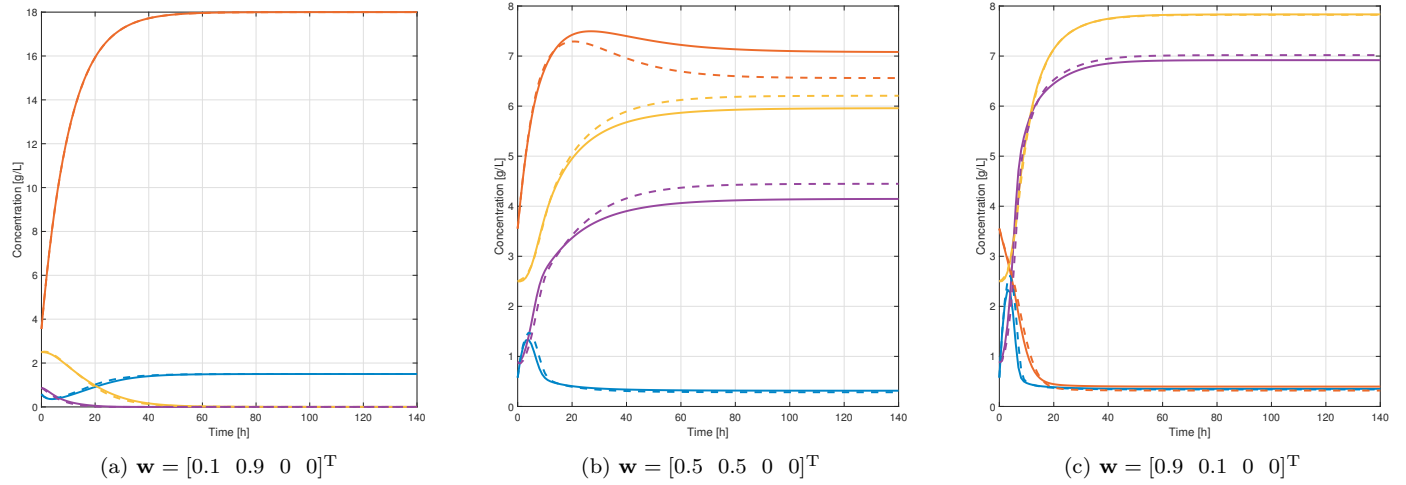


Fig. 2. Estimated macroscopic model (---) compared to the process modelled using metabolic network (—).

performance with the macroscopic model (NMPC-MA). In both cases, the prediction horizon is set to be 4 h, and the measurements are obtained every hour. For the case of NMPC-MN, the intracellular fluxes are estimated using an estimation horizon of 4 h.

The NMPC aims to track the product concentration. The set point is $C_{F,\text{set}} = 5.9580$ g/L. After 40 h, the set point is changed to $C_{F,\text{set}} = 7.8320$ g/L, and then brought back to the initial set point at 80 h.

Figure 3 depicts the fluxes estimated from the extracellular measurements using the NMHE framework described above. The estimated fluxes are in good agreement with the *real* fluxes. The discrepancy during the first few hours is when the full measurement horizon is unavailable for the NMHE estimation. In this period, the estimates are

obtained by the Unscented Kalman Filter. Once the full horizon is available for estimation, the UKF serves as the arrival cost estimate for the NMHE.

Figure 4a depicts the performance of the two NMPCs for tracking of the product F. The corresponding input profiles are depicted in Figure 4b. Note that for optimisation only the weight substrate A (w_1) is considered as the variable. The weight associated to the substrate E then is $w_2 = 1 - w_1$. Although both schemes perform well, there is a small offset when the set point is $C_{F,\text{set}} = 5.9580$ g/L. This offset is explained by the model-process mismatch highlighted in Figure 2. The mismatch is higher at $\mathbf{w} = [0.5 \ 0.5 \ 0 \ 0]$, than at other inputs. As the set point $C_{F,\text{set}} = 5.9580$ g/L requires the weights to be near 0.5, the plant-process mismatch is translated into an offset.

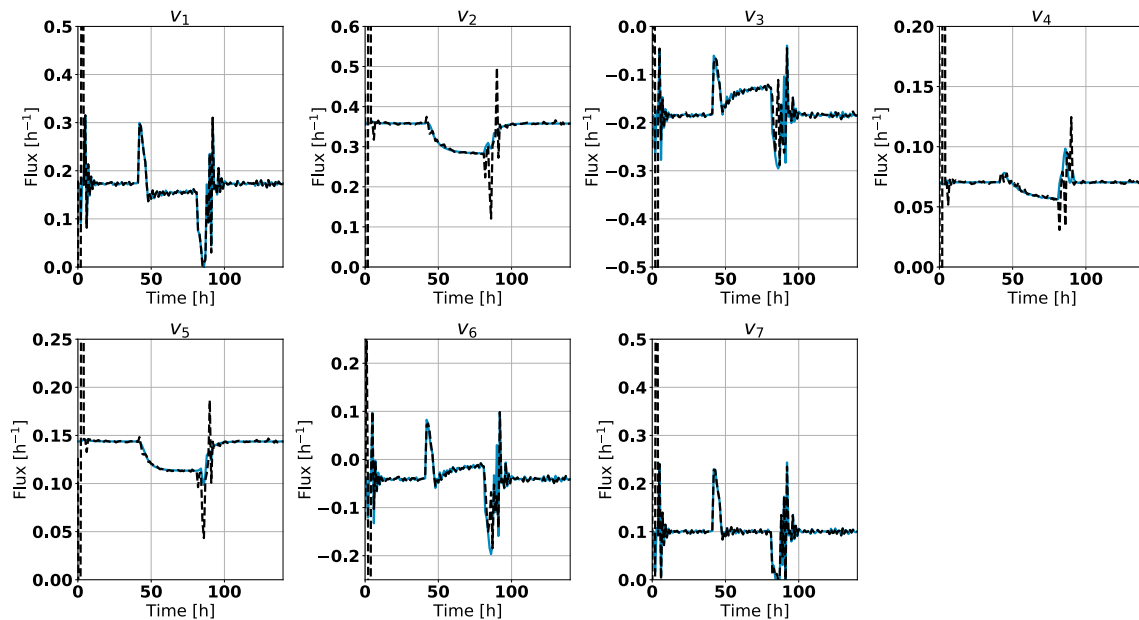
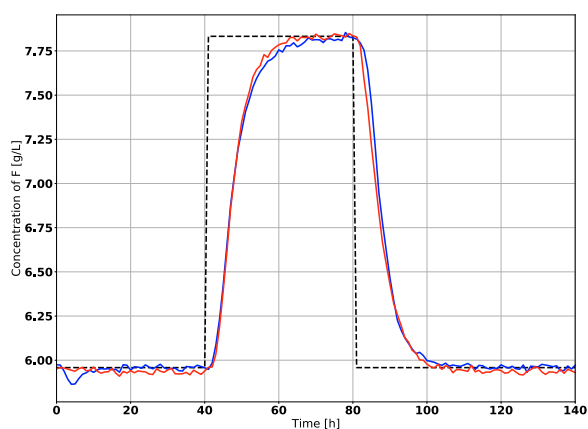
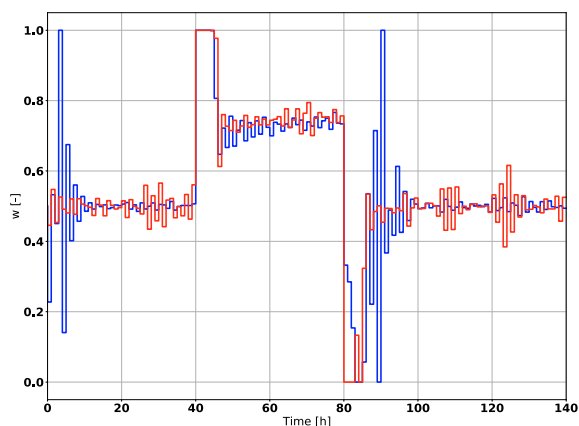


Fig. 3. Fluxes estimated by the NMHE (---) compared to the *real* fluxes (—).



(a) Tracking performance of NMPC-MN (blue line) and NMPC-MA (red line). The set point is depicted by the black dashed line



(b) The optimised control profile with NMPC-MN (blue line) and NMPC-MA (red line).

Fig. 4. NMPC using metabolic networks and macroscopic models

In this example, the improvement in the tracking performance by NMPC-MN is marginal in comparison to NMPC-MA. In our example, the macroscopic model provides a good fit to the data. This is partly due to a good choice of reaction kinetics. The kinetics (Eqs. (17 - 20)) resemble the *true* kinetics used in the process plant (Eqs. (11 - 13)) with the exception of u_4 . The choice of kinetics in a multi-substrate bioreactor model is not trivial, and a large amount of experimental effort needs to be dedicated when identifying proper kinetics. The use of metabolic networks removes this reliance on empirical kinetics and utilises cellular information. Furthermore, the case study considers a simplistic network. For more complex organisms, the difference in accuracy of metabolic network models and an identified macroscopic model might be much larger.

The median computational time per iteration for the NMPC-MN is 0.80 s and the median time for NMPC-MA is 0.03 s. The longest iteration for NMPC-MN is 1.18 s, while the longest iteration for NMPC-MA is 0.06 s. This difference is expected as the NMPC-MN problem to

be solved is significantly more complex than the one in NMPC-MA. Furthermore, NMPC-MN requires the solution of two optimisation problems rather than just one. In either case, the computational time is orders of magnitude slower than the measurement frequency of 1 h. Thus, both the NMPC schemes can be deployed in practice.

The computational complexity of NMPC-MN will scale with the size of the metabolic networks involved. A genome-scale model consists of thousands of metabolites, reactions, and fluxes. Given the lack of intracellular measurements, the system is often underdetermined. Thus, the large-scale models need to be reduced to provide identifiable fluxes and to manage the computational complexity.

4. CONCLUSIONS

This paper compares the metabolic network-based NMPC with a macroscopic model based NMPC. In the NMPC-MN, the intracellular fluxes are identified using the NMHE framework. The unknown fluxes are parameterised using the Haag flux model. For the macroscopic model, reaction kinetics are assumed to be known. In the example considered, the metabolic network-based NMPC provides slight improvement to the tracking performance as compared to the macroscopic model based NMPC. The improvement comes at the cost of increased computational complexity. Given the marginal gains, NMPC-MA would be the preferred scheme for optimisation and control if the macroscopic model is well identified. However, identifying the structure of the reaction kinetics, especially for multi-substrate processes, is not trivial. This limits the applicability of the macroscopic model to the range it was built on.

NMPC-MN on the other hand utilises the cellular information available via the stoichiometric matrix. However, incorporating a genome-scale metabolic network is computationally expensive and also leads to identifiability issues. Thus, the metabolic network to be used in the NMPC-NMHE framework needs to be reduced to ensure feasibility and identifiability. In future, we will evaluate various network reduction approaches to reduce the large scale models for use in such an NMPC-NMHE framework.

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